Lecture plan · for the lecturer

Sampling distributions and the bootstrap

Data science · University · Lecture 1

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Define a sampling distribution and SE.
- Use the bootstrap and know its limits.
- Sequence teaching simulation-first.

Activity: Resample before you derive

1. Warm-up (5 min): Ask students to write, alone, what a 95% confidence interval means; collect and sort the answers.
2. Simulate (20 min): Pairs draw 1,000 samples of n = 10 and n = 40 from a skewed population and plot the means.
3. Derive (15 min): Derive $\mathrm{Var}(\bar{x}) = \sigma^2/n$ on the board and check it against the two plots.
4. Bootstrap (15 min): Give one sample; students bootstrap the median and discuss when the method fails.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Which plot was narrower, and by how much?
- Why can't resampling fix a biased sample?
- How will you assess interpretation, not just calculation?

Exit ticket

- Write one correct sentence interpreting a 95% confidence interval.

Try it in your next lecture
Before showing any formula, give pairs one dataset and ask them to resample it 200 times and plot the means; then ask them to predict the plot for a sample four times larger.

Seminar sheet · for students

Sampling distributions and the bootstrap

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Name _____ Group _____ Date _____

1 **A study has a standard error of 4. How many times more data bring it down to 2?**

◯ Twice as much ◯ Four times as much ◯ Eight times as much

2 **Work out the standard errors.**

With σ = 12 and n = 36, SE = _____. With n = 144, SE = _____.

Word bank: 4 · 2 · 1 · 0.5

3 **When is the plain percentile bootstrap a poor choice? Choose all that apply.**

☐ Observations are strongly dependent over time ☐ Estimating the maximum of a distribution

☐ The sample was drawn only from volunteers ☐ Estimating the SE of a median from n = 200

4 **By the central limit theorem, the means of large samples from a skewed population pile up in a near-normal shape, while each sample itself stays skewed.**

◯ True ◯ False

5 **Match each student statement to the misconception behind it.**

"There's a 95% chance μ is in my interval."	• • • •	Resampling can't add information
"A bigger sample makes the data normal."	• • • •	Spread of an estimate versus spread of the data
"Bootstrapping a million times fixes a small biased sample."	• • • •	Probability attached to one fixed interval
"SE and SD are the same thing."	• • • •	Confusing data with the sampling distribution

Answers · for the lecturer

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1 **Four times as much**

SE falls like $1/\sqrt{n}$, so halving it needs four times the data.

2 **With $\sigma = 12$ and $n = 36$, SE = 2. With $n = 144$, SE = 1.**

$12/\sqrt{36} = 12/6 = 2$; quadrupling n to 144 gives $12/12 = 1$.

3 **Estimating the maximum of a distribution; The sample was drawn only from volunteers; Observations are strongly dependent over time**

Resampling can't create values beyond the sample's maximum, can't fix a biased sample, and breaks time dependence unless you resample blocks; a median from 200 cases is a standard good use.

4 **True**

The CLT is about the sampling distribution of the mean, not the data, which stay skewed.

5 **"There's a 95% chance μ is in my interval." — Probability attached to one fixed interval**
"A bigger sample makes the data normal." — Confusing data with the sampling distribution
"Bootstrapping a million times fixes a small biased sample." — Resampling can't add information
"SE and SD are the same thing." — Spread of an estimate versus spread of the data

Naming the misconception lets you write a question that diagnoses it, rather than just marking the answer wrong.

For you to think about

Where in your current module do students first meet a standard error? What would a simulation-first version of that session look like?